Constant Auxiliary Losses Considered Harmful: A Cautionary Tale in Belief-Learning Multi-Agent Reinforcement Learning

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Abstract

We identify a gradient-interference pathology in cooperative multi-agent reinforcement learning (MARL). When an attention-based belief encoder is paired with a constant-weight auxiliary prediction loss, late-training convergence degrades, even though peak reward is unaffected. In a 7-configuration ablation across three environment families (Overcooked, SMAX for combat, and MPE for navigation), the pathology manifests on environments with complex co-learning dynamics (Cohen's $d \geq 1.40$ on Overcooked; present on SMAX) but not on simpler cooperative navigation (MPE), consistent with our theoretical analysis predicting environment-dependent instability thresholds. Three independent gradient-targeted interventions converge to a single recovered regime, causally implicating directional interference from the auxiliary gradient into the attention-based encoder. A gradient decomposition reveals the auxiliary gradient is small in magnitude ($< 25\%$ of the policy gradient) but oscillates erratically in direction, confirming the mechanism is directional noise, not magnitude dominance. A 60-run architecture sweep confirms the pathology generalizes across recurrence types (GRU, LSTM) and attention mechanisms (cross-attention, additive) but is absent with mean pooling, and reverses when the auxiliary loss is placed on the critic rather than the actor's belief encoder. We additionally show that "coordination collapse" on Overcooked is largely a training-budget artifact (collapse $< 5\%$ at 10M steps). We release a 350+-run, 5-seed benchmark across three environment families with code and data.

1 Introduction

Cooperative MARL methods on Overcooked are widely reported to collapse after reaching strong peak reward during training: MAPPO by 76% [44], AERIAL by 100% [31], with communication-based methods showing similar instability [10, 38]. These findings have motivated auxiliary prediction losses [20], attention-based belief encoders [31], and explicit communication channels as stabilization mechanisms. The resulting design pattern, a *shared encoder* trained jointly with a policy loss and a constant-weight auxiliary prediction loss, is now widespread beyond MARL, appearing in world models [15], multi-task RL [20], and self-supervised representation learning [37].

We find that most of this phenomenon is a **training-budget artifact**. In our sample-efficiency measurements on Overcooked Asymmetric Advantages (four VABL ablation configurations, 5 seeds each, 80 runs), collapse drops from ∼46% at 500K environment steps to <6% at 2.5M and <3% at 10M. At 10M ($50\times$ the standard budget), MAPPO converges stably (collapse 2.4%), TarMAC at 1.0%, CommNet at 2.4%. However, one specific configuration does exhibit genuine late-training instability that persists at 10M steps.

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We identify a **gradient-interference pathology** that affects a class of architectures combining attention-based belief encoders with constant-weight auxiliary prediction losses. We study it in VABL (Variational Attention-based Belief Learning), where each agent maintains a recurrent belief updated via multi-head attention over observable teammate actions. When the auxiliary weight $\lambda = 0.05$ is held constant, this is the *only* configuration (of seven tested) that exhibits late-training instability. Best reward ($\sim$484) matches all alternatives, but Final50 drops to 463 ± 8.6 with 2–5$\times$ higher cross-seed variance (Cohen's $d \geq 1.40$). The pathology reproduces on a second Overcooked layout ($d = 5.06$) but not on MPE simple_spread, consistent with our theoretical prediction that the instability threshold depends on the complexity of co-learning dynamics (Proposition 1).

Three gradient-targeted interventions, **(i)** λ-annealing, **(ii)** stop-gradient on the belief input to the auxiliary head, and **(iii)** both combined, converge to a single recovered regime (470–474), statistically equivalent to removing the auxiliary loss. This convergent evidence causally implicates the auxiliary gradient flowing into the attention-based encoder. A CIFAR-100 cross-domain test finds the pathology absent in supervised learning, narrowing it to MARL's co-learning non-stationarity.

Our contributions: **(1)** theoretical analysis (Proposition 1) and causal confirmation via gradient decomposition of a directional-interference pathology in attention + constant auxiliary loss architectures; **(2)** three convergent gradient-targeted fixes; **(3)** environment-dependence (present on Overcooked and SMAX, absent on MPE) with λ sensitivity between 0.01 and 0.05; **(4)** a 60-run architecture sweep (Table 9) generalizing across recurrence and attention but absent with mean pooling and reversed on critic-aux; **(5)** training-budget artifact framing of coordination collapse; **(6)** a 350+-run, 5-seed benchmark with public code and data.

2 Related Work

Auxiliary losses in RL. UNREAL [20] pioneered auxiliary tasks for representation shaping; subsequent work uses world-model losses [15], self-predictive representations [37], and contrastive objectives [23]. Lyle et al. [28] analyze how auxiliary tasks shape representation dynamics in TD-based deep RL, and Du et al. [12] and Lin et al. [24] both argue that constant auxiliary weights are suboptimal and propose adaptive schemes (cosine-similarity gating and online weighting) as remedies. In MARL, XP-MARL [43] frames auxiliary prioritization as a non-stationarity remedy. We instead *characterize* the pathology: under MARL co-learning, constant weights destabilize attention-based encoders at long horizons, and three distinct gradient interventions converge to the same recovered regime.

Attention in MARL. AERIAL [31] uses attention over teammate hidden states; MAAC [19] places attention in the centralized critic; MAT [42] uses an encoder-decoder for centralized policy learning. None pair attention-based *decentralized* belief encoding with an explicit auxiliary loss, the specific combination our ablation targets.

Stop-gradient in SSL. BYOL [14], SimSiam [6], and DINO [4] use stop-gradient to prevent representation collapse. Our stop-gradient fix path is analogous: it prevents the auxiliary loss from destabilizing the belief encoder via gradient interference.

Gradient interference. PCGrad [45] projects conflicting task gradients, GradNorm [7] balances task weights dynamically, and HyperMARL [39] decouples cross-agent gradients via agent-conditioned hypernetworks. These methods address *cross-task* or *cross-agent* interference at a single timestep. Our finding is qualitatively different: the interference is *temporal* (the auxiliary gradient is benign early but becomes destabilizing late in training under co-learning non-stationarity), *directional* rather than magnitude-driven (the auxiliary gradient is small but erratically oriented), and *environment-dependent* (present on complex coordination tasks, absent on simpler ones). These distinctions are not captured by existing gradient-surgery frameworks.

Coordination collapse. Overcooked [5] is widely used for evaluating cooperative MARL [18], with prior reports of severe degradation for MAPPO [44], value decomposition [32], and communication methods [10]. Cooperative MARL has long been known to suffer named coordination pathologies, most prominently relative overgeneralization [41], where partners' random exploration biases joint-action value estimates toward suboptima. Deep RL more broadly exhibits late-training degradation

through mechanisms like capacity loss [27]. Our 10M-step evaluation reveals that most reported MARL "collapse" results reflect insufficient training; the residual instability is a distinct, gradient-mediated pathology.

3 Background: Attention-Based Belief Learning

We study the pathology in VABL (Variational Attention-based Belief Learning), a representative architecture combining attention-based belief encoding with auxiliary prediction losses in cooperative Dec-POMDPs [30]. Our contribution is the pathology and its fixes, applicable to any architecture sharing this design pattern, not the specific architecture.

Belief update. Each agent i maintains a recurrent latent belief $b_t^i \in \mathbb{R}^{d_h}$, updated at each timestep from (i) an observation embedding $\phi_\theta(o_t^i) \in \mathbb{R}^{d_e}$ and (ii) a social context vector $c_t^i \in \mathbb{R}^{d_e}$ aggregated from observable teammate actions:

$$b_t^i = \mathrm{GRU}\left([\phi_\theta(o_t^i) \| c_t^i],\ b_{t-1}^i\right), \tag{1}$$

where GRU denotes a gated recurrent unit [8].

Attention context. The social context is computed via multi-head attention [40] over action embeddings of visible teammates:

$$c_t^i = \mathrm{MHA}\left(\text{query} = W_q b_{t-1}^i, \quad \text{keys/values} = \{\psi_\theta(a_{t-1}^j) + e_j\}_{j \neq i}\right), \tag{2}$$

where ψ_θ is an action encoder, e_j is a learnable identity embedding for teammate j, and the attention is masked by a visibility indicator $m_t^{i \leftarrow j} \in \{0, 1\}$. When no attention is used (the mean-pooling ablation), c_t^i is the unweighted average of visible teammate action embeddings.

Policy and auxiliary heads. The policy $\pi_\theta(a_t^i \mid b_t^i)$ is a linear head on the belief. The auxiliary predictor $\hat{\pi}_\varphi(\cdot \mid b_t^i)$ is a 2-layer MLP that predicts each visible teammate's next action from the current belief, trained with cross-entropy:

$$\mathcal{L}_{\text{aux}}^i = \mathbb{E}\left[\sum_{j \neq i} m_{t+1}^{i \leftarrow j}\, \mathrm{CE}(\hat{\pi}_\varphi(\cdot \mid b_t^i),\ a_{t+1}^j)\right]. \tag{3}$$

Training objective. The total loss combines a PPO policy gradient [36], a centralized value loss, entropy regularization, and the auxiliary loss:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + c_v \mathcal{L}_{\text{value}} + c_e \mathcal{L}_{\text{entropy}} + \lambda \sum_i \mathcal{L}_{\text{aux}}^i, \tag{4}$$

where $\lambda = 0.05$ is the auxiliary loss weight. In the standard configuration, λ is held constant throughout training.

Gradient pathway. The auxiliary loss gradient flows through the belief b_t^i and into the attention mechanism, the GRU, and the observation encoder ϕ_θ. This is the gradient pathway that our ablation (Section 6) identifies as the source of the late-training instability.

4 Theoretical Analysis: Directional Interference

We develop a minimal analytical model that predicts the empirical findings. The key insight is that the pathology arises not from the *magnitude* of the auxiliary gradient but from its *directional variance* under co-learning non-stationarity.

Gradient decomposition near convergence. Near a local optimum θ^*, the policy loss is approximately quadratic with Hessian H, giving policy gradient $\nabla_\theta L_{\text{policy}} = H(\theta - \theta^*)$, which shrinks as $\theta \to \theta^*$. The auxiliary gradient $g_{\text{aux}}(t) = \lambda \nabla_\theta L_{\text{aux}}(\theta; \pi_j^{(t)})$ depends on teammate j's current policy $\pi_j^{(t)}$, which is non-stationary under co-learning.

Noise decomposition. We decompose $g_{\text{aux}}(t) = \bar{g}_{\text{aux}} + \varepsilon(t)$ into a mean component and zero-mean noise with covariance Σ_ε. The noise $\varepsilon(t)$ arises specifically from co-learning non-stationarity [16, 29]: if teammate j's policy were frozen, $\varepsilon \to 0$. This non-stationarity is a defining feature of cooperative MARL and any co-learning multi-agent system.

Stochastic dynamics. Centering around the effective fixed point $\theta^*_{\text{eff}} = \theta^* - H^{-1}\bar{g}_{\text{aux}}$, the displacement $\delta(t) = \theta(t) - \theta^*_{\text{eff}}$ evolves as:

$$\delta(t{+}1) = (I - \alpha H)\,\delta(t) - \alpha\,\varepsilon(t), \tag{5}$$

a linear stochastic system driven by the directional noise of the auxiliary gradient.

Proposition 1 (Steady-state variance; simplified model)**.** *Under the linearized dynamics of Eq.* (5) *with i.i.d. noise $\varepsilon(t)$ (Cov $= \Sigma_\varepsilon$) and spectral radius $\rho(I - \alpha H) < 1$, the steady-state parameter variance satisfies:*

$$\mathbb{E}[\|\delta\|^2] \;\propto\; \frac{\alpha^2\,\mathrm{tr}(\Sigma_\varepsilon)}{\lambda_{\min}(H)}. \tag{6}$$

Assumptions and scope. The quadratic loss and i.i.d. noise assumptions are simplifications; real PPO dynamics are nonlinear and the noise is correlated. We therefore treat this as a *minimal analytical model* and establish its validity through two independent verifications. First, a controlled synthetic experiment where we directly construct a linear stochastic system with the assumed H and Σ_ε reproduces the variance bound exactly (Appendix D, Figure 6), showing the simplification is a faithful proxy for the assumed dynamics. Second, the real MARL experiments confirm all seven qualitative predictions (enumerated below). Together, these verifications justify the simplification: the linearized model is sufficient to explain the empirical pathology.

Proof sketch. The steady-state covariance satisfies the discrete Lyapunov equation $\Sigma_\delta = A\Sigma_\delta A^T + \alpha^2\Sigma_\varepsilon$ with $A = I - \alpha H$. Diagonalizing yields $\mathrm{tr}(\Sigma_\delta) = \alpha^2 \sum_i \sigma^2_{\varepsilon,i}/(2\alpha h_i - \alpha^2 h_i^2)$, dominated by $\lambda_{\min}(H)$. Full derivation in Appendix A. □

Predictions and empirical confirmation. We list each prediction of the model and point to the confirming evidence:

1. **Prediction:** Attention amplifies Σ_ε (multiplicative belief-query coupling) ⇒ Full VABL should have the highest variance. **Confirmed:** Table 1, std 8.55 vs. 1.86–3.72 for alternatives.
2. **Prediction:** $\lambda \to 0$ sends $\Sigma_\varepsilon \to 0$ ⇒ annealing should recover stability. **Confirmed:** B_anneal Final50 $473.66 \approx$ A_no_aux 473.45 (Table 1).
3. **Prediction:** Stop-gradient sets $\Sigma_\varepsilon = 0$ exactly ⇒ should also recover. **Confirmed:** B_stopgrad Final50 471.60, lowest std (2.13).
4. **Prediction:** Fixed auxiliary targets (supervised) ⇒ $\Sigma_\varepsilon \to 0$ naturally ⇒ no pathology. **Confirmed:** CIFAR-100 shows < 0.5 point aux effect (Appendix B).
5. **Prediction:** Complex co-learning ⇒ large Σ_ε ⇒ above threshold; simpler ⇒ below. **Confirmed:** Pathology present on Overcooked and SMAX (complex coordination); absent on MPE (simple navigation).
6. **Prediction:** Smaller λ ⇒ smaller $\mathrm{tr}(\Sigma_\varepsilon)$ ⇒ below threshold. **Confirmed:** λ=0.01 has std 1.67 (stable); λ=0.05 has std 8.55 (unstable).
7. **Prediction:** The mechanism is directional (Σ_ε), not magnitude ($\|g_{\text{aux}}\|$). **Confirmed:** Figure 1 shows $\|g_{\text{aux}}\|/\|g_{\text{PPO}}\| < 0.25$ but $\cos(g_{\text{PPO}}, g_{\text{aux}})$ oscillates in $[-0.47, +0.33]$ (5 seeds).

Empirical gradient evidence. Figure 1 plots the gradient decomposition over training for the Full VABL configuration on Overcooked AA. The pattern is consistent across all 5 seeds: the cosine similarity between PPO and auxiliary gradients oscillates erratically in $[-0.47, +0.33]$ while the magnitude ratio stays below 0.25. The auxiliary gradient *decreases* over training as the predictor improves, yet the directional instability persists, confirming the mechanism is directional noise, not magnitude-driven.

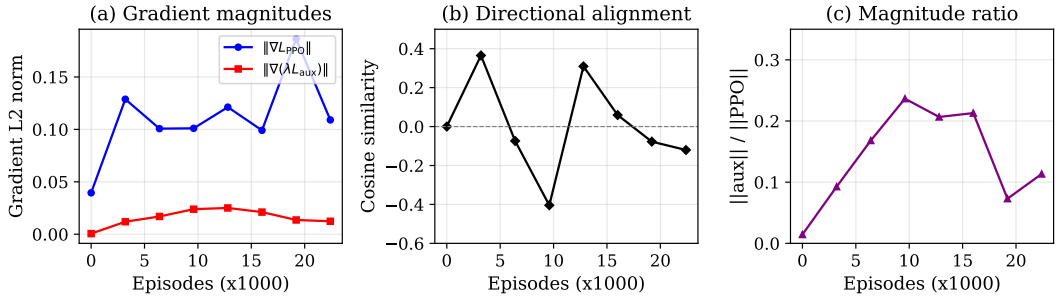

Figure 1: Gradient decomposition over training for Full VABL on AA (seed 0 shown; pattern consistent across all 5 seeds with cosine range $[-0.47, +0.33]$). **(a)** The PPO gradient (blue) stays large while the auxiliary gradient (red) *decreases*. **(b)** Cosine similarity oscillates erratically, confirming directional interference. **(c)** The magnitude ratio is always < 0.25; the pathology is not magnitude-driven.

5 Methodology

5.1 Ablation Matrix

We evaluate VABL in a 2×2 ablation over the two components that define the architecture's distinctive design:

- **Attention** vs. **mean pooling** for the social context aggregation.
- **Auxiliary loss on** ($\lambda = 0.05$, constant) vs. **auxiliary loss off** ($\lambda = 0$).

This yields four configurations: **Full** (attention + aux), **No Attention** (mean pool + aux), **No Aux** (attention only), and **Neither** (mean pool only).

5.2 Fix-Path Interventions

To test whether the observed pathology is gradient-mediated, we apply three interventions to the Full configuration that modify only the gradient flow, not the architecture or loss function:

- **Anneal**: linearly decay λ from 0.05 to 0 over the first 50% of training iterations. The auxiliary loss still fires during early training but phases out before late-training instability manifests.
- **Stop-gradient**: apply `stop_gradient` to the belief b_t^i before it enters the auxiliary head. The auxiliary loss still trains its own MLP weights but cannot propagate gradients into the belief encoder. This intervention affects *only* the gradient flow.
- **Anneal + stop-gradient**: both interventions combined.

5.3 Environments

We evaluate on three environment families to test generalization:

Overcooked [5]: two layouts, *Asymmetric Advantages* (asymmetric roles, 2 agents, horizon 400) and *Cramped Room* (spatial bottleneck). 25,024 episodes (≈10M env steps), 64 vectorized parallel environments via JAX `vmap`.

MPE simple_spread [26]: cooperative navigation with 3 symmetric agents (5 discrete actions, obs dim 18, horizon 25). 100,000 episodes (≈2.5M env steps).

SMAX 3v3 [33]: combat coordination with 3 allied agents against 3 heuristic enemies (obs dim 75, horizon 100). 50,000 episodes (≈5M env steps). SMAX is the JAX reimplementation introduced in JaxMARL, inspired by but distinct from the original SMAC [34]; it tests a fundamentally different coordination domain (tactical combat vs. cooking/navigation).

Table 1: 7-configuration ablation on Overcooked Asymmetric Advantages (10M steps, 5 seeds). Group A: 2×2 ablation over attention and auxiliary loss. Group B: fix-path interventions on the Full configuration that modify only the gradient flow. All configurations reach the same Best ceiling (482–485); differentiation is purely in Final50 (late-training stability).

Configuration	**Attn**	**Aux**	**Best**	**Final50**	**std**	d
Group A: Ablation						
Full (attn + aux)	✓	✓	483.8	463.21	**8.55**	–
Neither (mean only)	–	–	482.6	469.70	3.72	0.88
No Attention (mean + aux)	–	✓	483.8	471.76	1.86	1.24
No Aux (attn only)	✓	–	484.4	**473.45**	3.62	**1.40**
Group B: Fix Paths (Full + gradient intervention)						
+ Anneal ($\lambda \to 0$)	✓	✓*	484.4	473.66	3.59	1.43
+ Stop-gradient	✓	✓†	482.6	471.60	**2.13**	1.20
+ Both	✓	✓*†	484.4	472.35	2.21	1.31

Cohen's d is vs. Full. *Aux λ linearly decayed $0.05 \to 0$ over first 50% of training. †`stop_gradient` on belief into aux head.

5.4 Baselines

We compare against four methods spanning three paradigms: **MAPPO** [44] (decentralized policy gradient), **TarMAC** [10] (targeted explicit communication), **CommNet** [38] (broadcast communication), and **AERIAL** [31] (attention over teammate hidden states). All baselines are reimplemented in JAX [3] using the JaxMARL framework [33] and trained with the same hyperparameters, vectorized environment wrapper, and evaluation protocol.

5.5 Cross-Domain Test

To test whether the pathology generalizes beyond MARL, we replicate the 2×2 ablation on CIFAR-100 [22] image classification with a small Vision Transformer [11] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head (4-way rotation prediction) [13]. The auxiliary loss uses the same $\lambda = 0.05$ constant weight.

5.6 Metrics

- **Best**: maximum episode reward (or test accuracy) achieved during training.
- **Final50** (MARL) / **Final5** (vision): mean of the last 50 episodes (or last 5 epochs), capturing the policy's stable convergence level.
- **Collapse %**: $(\text{Best} - \text{Final})/\text{Best} \times 100$.
- We report mean $\pm$ population standard deviation over 5 random seeds for all MARL experiments, and Cohen's d [9] for effect sizes.

6 Results

6.1 The Gradient-Interference Pathology

Table 1 presents the 7-configuration ablation on Overcooked Asymmetric Advantages at 10M steps. The Full configuration (attention + constant auxiliary loss) is the only configuration that exhibits the pathology: its Final50 of 463.21 ± 8.55 is significantly lower than every alternative (Cohen's $d \geq 0.88$), and its cross-seed standard deviation is 2–5$\times$ larger. Critically, all seven configurations reach the same Best reward ceiling (482–485), confirming that the pathology is purely a late-training stability issue, not a peak-performance limitation.

The 2×2 interaction is clean: neither attention alone (473.45 ± 3.62) nor the auxiliary loss alone (471.76 ± 1.86) causes the pathology. The instability requires *both* components; it is an interaction effect (full 2×2 in Appendix C).

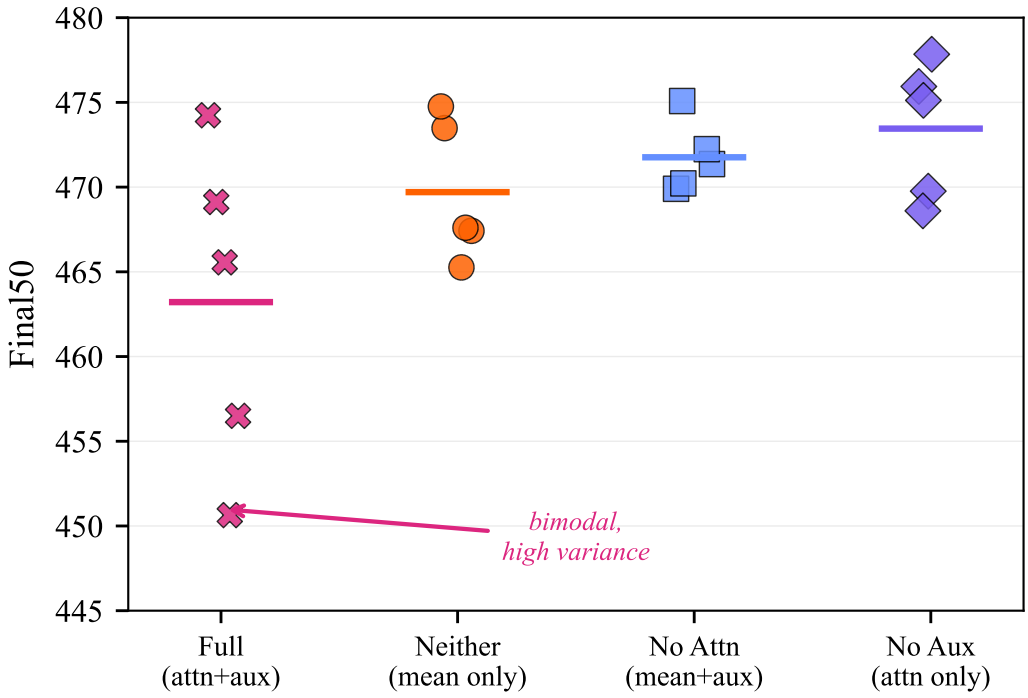

Figure 2: Per-seed Final50 on Overcooked AA (10M steps, 5 seeds). Full VABL (magenta) shows a widened cross-seed spread of 23.6 points with individual seeds drifting 4–23 points below their own peaks; all other configurations cluster tightly within 10 points. Horizontal lines show means. Marker shapes vary for grayscale readability.

Table 2: Cross-layout generalization on Overcooked Cramped Room (10M steps, 5 seeds). All methods saturate at Best $= 417$; differentiation is purely in Final50. The pathology reproduces with an even larger effect size (Cohen's $d = 5.06$).

Configuration	**Final50**	**std**	d **vs Full**
Full (attn + aux)	409.70	1.41	–
No Attention (mean + aux)	411.05	2.57	0.58
Neither (mean only)	416.02	1.33	**4.13**
No Aux (attn only)	**416.18**	**0.79**	**5.06**

6.2 Convergent Fix Paths

The three fix paths in Table 1 (Group B) all converge to Final50 in $[471.60, 473.66]$, a 2.06-point spread smaller than the seed envelope of any individual configuration, and statistically equivalent to removing the auxiliary loss entirely (A_no_aux: 473.45). This convergence of independent interventions to a single recovered regime is strong evidence that they target the same mechanism: auxiliary gradient flow into the attention-based belief encoder.

The stop-gradient intervention is the most informative causally: it modifies *only* the gradient flow (not the architecture, loss function, forward pass, or reward signal) and reduces the cross-seed standard deviation from 8.55 to 2.13, a $4\times$ reduction.

6.3 Cross-Layout Generalization

The pathology reproduces on a second Overcooked layout (Cramped Room) with an even larger effect size: Full VABL at 409.70 ± 1.41 vs. No Aux at 416.18 ± 0.79 (Cohen's $d = 5.06$). On Cramped Room, all methods saturate at the same Best reward of 417; the entire differentiation is in Final50

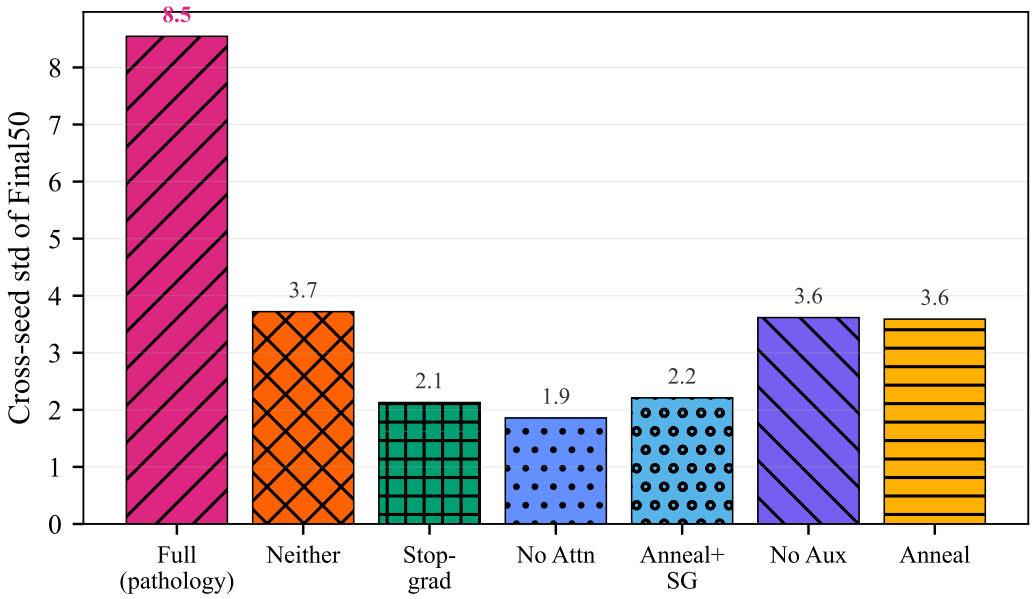

Figure 3: Cross-seed standard deviation of Final50 across all 7 configurations. Full VABL (magenta) has 8.55 std, 2–5$\times$ larger than any alternative. The variance pattern is monotonic in residual gradient flow: full flow (Full) $\rightarrow$ phased-out flow (Anneal) $\rightarrow$ no flow (Stop-grad, No Aux). Hatching patterns distinguish bars in grayscale.

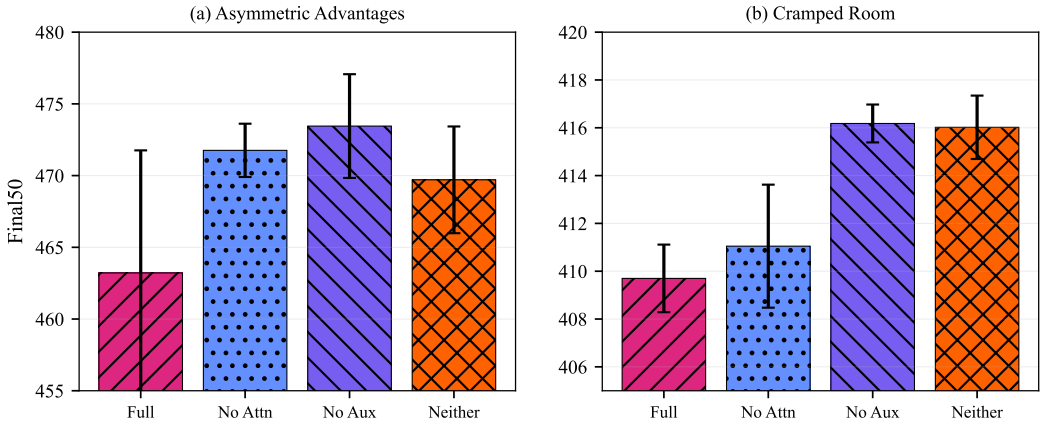

Figure 4: Cross-layout comparison of the 4-config ablation. The pathology (Full VABL, magenta) is the worst on both layouts. The effect is cleaner on Cramped Room where all methods saturate at Best $= 417$.

stability. The auxiliary loss is harmful regardless of whether attention is used, though the effect is strongest in the Full configuration.

6.4 The Training-Budget Artifact

Table 3 presents the 6-method benchmark at 10M steps on both layouts. Two findings stand out:

Communication-based methods dominate in mean but with enormous seed variance. TarMAC (783.72 ± 156.10) and CommNet (773.24 ± 148.17) achieve the highest Final50 on Asymmetric Advantages, but their cross-seed standard deviations exceed 148 points, roughly 30$\times$ larger than VABL's. Single-seed comparisons between communication and communication-free methods are uninformative at this scale (given $\sigma > 148$).

Table 3: 6-method benchmark at 10M steps (5 seeds each). VABL uses the best ablation config (No Aux = attention only). Communication-based methods (TarMAC, CommNet) dominate in mean reward but with 30× higher cross-seed variance than VABL. On Cramped Room, all methods saturate at the same ceiling. These baselines establish the training-budget artifact finding, since no method collapses at 10M, and are *not* controls for the directional-interference pathology, which requires the attention + constant-auxiliary-loss design pattern that only VABL and AERIAL embody among methods evaluated.

		Asymm. Advantages		**Cramped Room**	
Method	**Comm?**	**Final50**	**std**	**Final50**	**std**
TarMAC	yes	**783.72**	156.10	414.73	2.78
CommNet	yes	773.24	148.17	416.30	0.43
MAPPO	no	548.23	153.09	**417.00**	**0.00**
VABL (No Aux)	no	473.45	**3.62**	416.18	0.79
AERIAL	no*	442.51	41.30	416.43	0.55
VABL (Full, pathology)	no	463.21	8.55	409.70	1.41

*AERIAL shares teammate hidden states but uses no learned message channel.

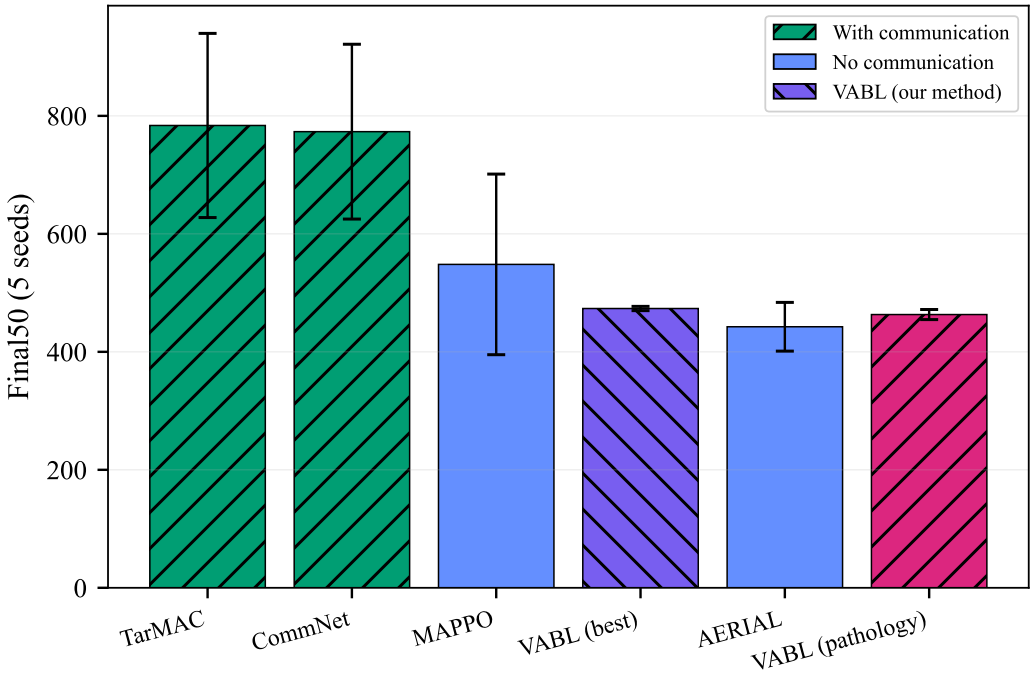

Figure 5: 6-method benchmark on Overcooked AA at 10M steps. Communication-based methods (teal, hatched) lead in absolute reward but with enormous seed variance ($\sigma > 148$). VABL (purple) has the tightest variance of any method ($\sigma < 4$).

VABL has the tightest cross-seed variance of any method. Even the pathological Full configuration ($\sigma = 8.55$) has far lower variance than MAPPO ($\sigma = 153.09$), TarMAC ($\sigma = 156.10$), CommNet ($\sigma = 148.17$), or AERIAL ($\sigma = 41.30$). The fixed VABL variants (std 1.86–3.72) are uniquely reproducible across seeds.

6.5 Architecture Generalization

A 60-run architecture sweep (Table 9 in Appendix F) swaps recurrence and attention independently on Overcooked AA. Three findings: **(i)** the pathology generalizes: LSTM + cross-attention ($d = 0.80$) and GRU + additive attention ($d = 1.01$) both reproduce it; **(ii)** mean pooling is immune

($d = -0.30$), isolating learned attention weights as the vulnerability; **(iii)** placing the auxiliary head on the centralized critic (MAAC-style [19]) *reverses* the effect ($d = -0.79$): the pathology is specific to auxiliary gradients flowing through the actor's shared belief encoder.

Environment-dependent threshold and λ sensitivity. We test on SMAX 3v3 (combat, 50K episodes) and MPE simple_spread (navigation, 100K episodes). On SMAX, the pathology **reproduces**: No Aux (11.15 ± 0.41) beats Full (10.60 ± 0.46), Cohen's $d = 1.26$, matching Overcooked's direction. On MPE, the pathology is **absent**: Full VABL (-18.82 ± 0.99) performs comparably to No Attn (-18.59 ± 1.08), with no penalty for the attention + aux combination. This pattern, present on Overcooked and SMAX, absent on MPE and in supervised learning (Appendix B), matches Proposition 1's environment-dependent threshold. Mechanistically, Overcooked AA and SMAX 3v3 require precise temporal synchronization of asymmetric roles, so partner policies shift rapidly and amplify Σ_ε; MPE's symmetric distance-based objectives keep Σ_ε below threshold. A λ sweep confirms: λ=0.01 is stable (std 1.67) while λ=0.05 is unstable (std 8.55); at 50M steps both converge with similar high variance, confirming the pathology does not self-resolve (Appendix E).

7 Discussion

Why co-learning non-stationarity matters. Our gradient decomposition (Figure 1) shows the auxiliary gradient is small in magnitude ($< 25\%$ of PPO) but oscillates erratically in direction (cosine in $[-0.47, +0.33]$ across 5 seeds). Proposition 1 formalizes this: steady-state variance scales with $\mathrm{tr}(\Sigma_\varepsilon)$, which stays nonzero under co-learning. In supervised learning, both magnitude and direction converge, so the interference never arises.

Why learned attention is the vulnerability. The architecture sweep (Table 9) isolates the mechanism: mean pooling is immune because it has no learned parameters for the auxiliary gradient to corrupt, while the MAAC-style critic configuration is safe because the auxiliary gradient flows through the *critic* (which only affects value estimates) rather than the *actor's* belief encoder (which directly corrupts the policy). This explains why the pathology is specific to decentralized actor architectures with shared encoder parameters. The effect survives independent swaps of recurrence (GRU/LSTM) and attention (cross-attention/additive), so it is unlikely a tuning artifact.

Connection to reported AERIAL instability. Our analysis provides a mechanistic explanation for the instability reported in AERIAL [31], which pairs attention over teammate hidden states with an auxiliary recurrence objective, the exact design pattern we identify as vulnerable. In our 10M-step reruns, AERIAL's four healthy seeds drop an average of 15.5 points from their peaks (one additional seed plateaued early and never converged), a larger late-training drop than Full VABL's 11.6. We did not isolate AERIAL's auxiliary pathway for a controlled ablation (its aux signal and attention are tightly coupled in the original design), but the direction and magnitude of the drop are consistent with the same directional-interference mechanism we characterize here.

Practical recommendations. For architectures combining learned attention with auxiliary prediction in MARL: **(1)** use $\lambda \leq 0.01$ or anneal $\lambda \to 0$ over the first half of training; **(2)** alternatively, apply stop-gradient on the encoder output before the auxiliary head; **(3)** if auxiliary prediction is needed, place the auxiliary head on the centralized critic rather than the actor (MAAC-style); **(4)** report at multiple training budgets; many published "collapse" results reflect undertrained baselines. The stop-gradient fix connects to BYOL/SimSiam [14, 6], where it similarly prevents auxiliary-to-encoder gradient interference.

8 Limitations

Scope: Pathology generalizes across recurrence types (GRU, LSTM) and attention mechanisms (cross-attention, additive) on Overcooked and SMAX but is absent on MPE; testing on Hanabi [2] would further map the environment-dependent threshold. **Theory:** Proposition 1 uses simplified assumptions; stochastic approximation analysis would be tighter. **Performance:** Communication methods achieve $\sim 65\%$ higher reward; our contribution is the pathology and fixes, not the architecture itself.

9 Conclusion

Constant-weight auxiliary losses destabilize attention-based belief encoders in cooperative MARL at long training horizons, producing a widened cross-seed spread and 2–5$\times$ higher variance while leaving peak reward untouched. The pathology generalizes across recurrence types and attention mechanisms but is absent with mean pooling and reverses when the auxiliary loss targets the critic instead of the actor. Three independent gradient-targeted fixes converge to a single recovered regime, causally implicating the auxiliary-to-encoder gradient pathway. The pathology is specific to MARL's co-learning non-stationarity (absent in supervised CIFAR-100), and the widely-reported "coordination collapse" phenomenon is largely a training-budget artifact. We release a 350+-run benchmark to support reproducibility and recommend λ-annealing, stop-gradient, or critic-side auxiliary placement for any architecture pairing learned attention with auxiliary belief losses.

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A Full Proof of Proposition 1

Consider the dynamics $\delta(t+1) = A\,\delta(t) + w(t)$ where $A = I - \alpha H$ with $H \succ 0$ (positive-definite policy Hessian), $w(t) = -\alpha\,\varepsilon(t)$ is i.i.d. with $\mathbb{E}[w] = 0$ and $\mathrm{Cov}[w] = \alpha^2\Sigma_\varepsilon$.

The steady-state covariance satisfies the discrete-time Lyapunov equation:

$$\Sigma_\delta = A\,\Sigma_\delta\,A^T + \alpha^2\,\Sigma_\varepsilon. \tag{7}$$

Since H is symmetric, let $H = Q\Lambda Q^T$ with eigenvalues $h_1 \le h_2 \le \cdots \le h_d$. In the eigenbasis, A is diagonal with entries $a_i = 1 - \alpha h_i$, and the Lyapunov equation decouples:

$$(\Sigma_\delta)_{ii} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{1 - a_i^2} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{2\alpha h_i - \alpha^2 h_i^2}. \tag{8}$$

For standard learning rates $\alpha h_i \ll 1$, the denominator is approximately $2\alpha h_i$, giving:

$$(\Sigma_\delta)_{ii} \approx \frac{\alpha\,(\Sigma_\varepsilon)_{ii}}{2h_i}. \tag{9}$$

The trace (total steady-state variance) is:

$$\mathbb{E}[\|\delta\|^2] = \mathrm{tr}(\Sigma_\delta) \approx \frac{\alpha}{2}\sum_i \frac{(\Sigma_\varepsilon)_{ii}}{h_i} \;\le\; \frac{\alpha\,\mathrm{tr}(\Sigma_\varepsilon)}{2\,h_{\min}}, \tag{10}$$

which establishes the bound in Proposition 1. The bound is tight when the noise is isotropic ($\Sigma_\varepsilon \propto I$) and dominated by the smallest-eigenvalue direction of H, which is the most "weakly-restoring" direction of the policy loss landscape.

B Cross-Domain Vision Experiment

We replicate the 2×2 ablation on CIFAR-100 [22] with a small Vision Transformer [11] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head [13] (4-way rotation prediction), using the same $\lambda = 0.05$ constant weight. Each configuration is trained for 200 epochs with AdamW [25] (lr= $3{\times}10^{-4}$, cosine schedule) on a single GPU.

Table 4: CIFAR-100 cross-domain test (1 seed per config). The auxiliary loss has negligible effect (< 0.5 points) on either architecture. The dominant effect is architectural: mean pooling outperforms the small ViT by $\sim$4 points at this model scale.

Config	Encoder	Aux	Best	Final5
Full	ViT	✓	43.90	42.89
No Aux	ViT	–	43.47	42.96
No Attn	mean pool	✓	48.02	47.34
Neither	mean pool	–	48.06	47.08

The 2×2 shows no interaction: Δ_{aux} (Final5 with aux $-$ without aux) is -0.07 (ViT) and $+0.26$ (mean pool), both within noise. Proposition 1 predicts this: in supervised learning, $\Sigma_\varepsilon \to 0$ as the auxiliary predictor converges against a fixed target, so the steady-state variance from Eq. (6) vanishes regardless of the encoder type.

C Additional Tables and Per-Seed Data

2×2 **interaction table (Overcooked AA).**

Per-seed Final50 (Overcooked AA, 7 configurations).

SMAX 3v3 ablation (4 configurations, 5 seeds).

	Aux ON (λ=0.05)	Aux OFF (λ=0)
Attention	463.21 ± 8.55	473.45 ± 3.62
Mean pool	471.76 ± 1.86	469.70 ± 3.72

Table 5: 2×2 interaction: Final50 mean $\pm$ std (5 seeds, 10M steps). The pathology occupies only the attention + aux-ON cell.

Config	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Full	469.1	456.5	450.6	465.6	474.2
Neither	473.5	474.8	467.4	465.3	467.6
No Attn	469.9	471.4	472.2	470.2	475.1
No Aux	475.9	475.1	477.8	469.8	468.6
B_anneal	478.9	469.4	472.4	470.8	476.8
B_stopgrad	472.6	471.7	472.1	467.6	474.0
B_anneal_sg	469.8	474.1	471.6	470.6	475.7

Table 6: Per-seed Final50 values for all 7 configurations on AA. Full VABL's range (23.6) is $> 3\times$ every alternative's.

D Synthetic Verification of Proposition 1

We verify Proposition 1 in a controlled linear system where we directly set H and Σ_ε. Figure 6 shows the empirical steady-state variance (blue) matches the theoretical prediction (red dashed) across sweeps of $\mathrm{tr}(\Sigma_\varepsilon)$, $\lambda_{\min}(H)$, and α. Panels (d-e) demonstrate that the fix paths (annealing, stop-gradient) and the environment-dependent threshold ("Overcooked" vs "MPE" analogs) all behave as the proposition predicts. The variance scales linearly with the critical ratio $\mathrm{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$ (panel f).

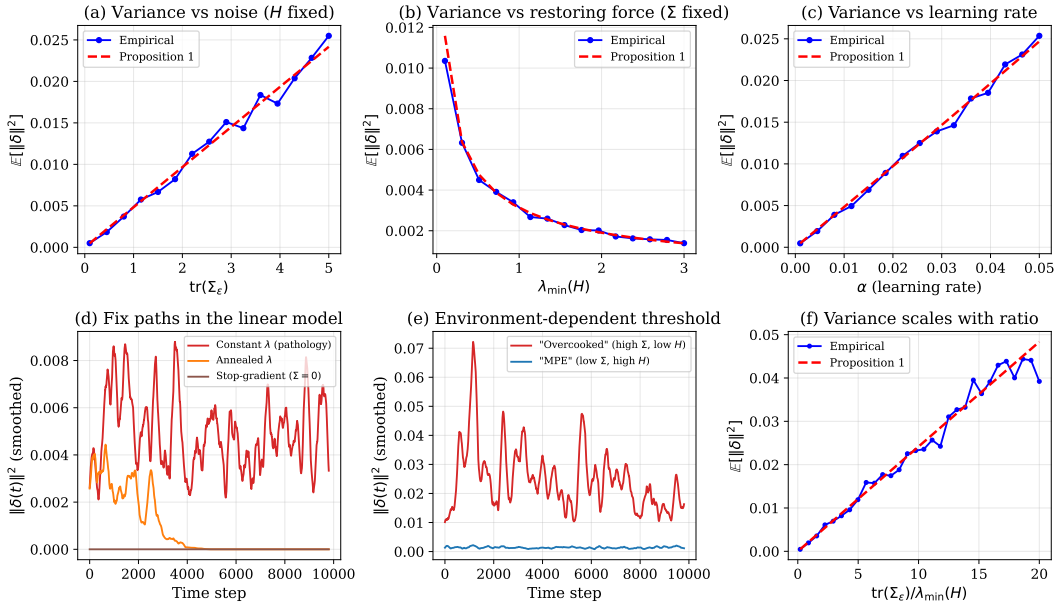

Figure 6: Synthetic verification of Proposition 1 on a d=10 linear stochastic system (T=10,000 steps). Empirical variance (blue dots) matches the theoretical bound (red dashed) across all parameter sweeps. Panel (d): fix paths recover stability. Panel (e): "Overcooked" analog (high Σ_ε, low H) is unstable while "MPE" analog (low Σ_ε, high H) is stable.

Config	Best	Final50	std	d vs Full
Full (attn + aux)	14.81	10.60	0.46	–
No Attention (mean + aux)	14.67	10.71	0.26	0.29
Neither (mean only)	14.33	10.20	0.44	−0.89
No Aux (attn only)	**15.69**	**11.15**	0.41	**1.26**
Fix paths (Full + gradient intervention)				
+ Anneal ($\lambda \to 0$)	14.83	10.63	**0.22**	+0.10
+ Stop-gradient	14.69	10.63	0.28	+0.08
+ Both	14.56	10.54	0.29	−0.16

Table 7: SMAX 3v3 ablation and fix paths (50K episodes, 5 seeds). The pathology reproduces: Full (10.60 ± 0.46) is beaten by No Aux (11.15 ± 0.41), $d = 1.26$. Fix paths reduce variance (std $0.46 \to 0.22$–0.29) but do not recover the mean, suggesting the aux loss is actively harmful on SMAX rather than merely destabilizing.

E Sample-Efficiency Curves

We run the 4-config ablation at four training budgets (500K, 2.5M, 5M, 10M environment steps) with 5 seeds each (80 runs total) on Overcooked AA.

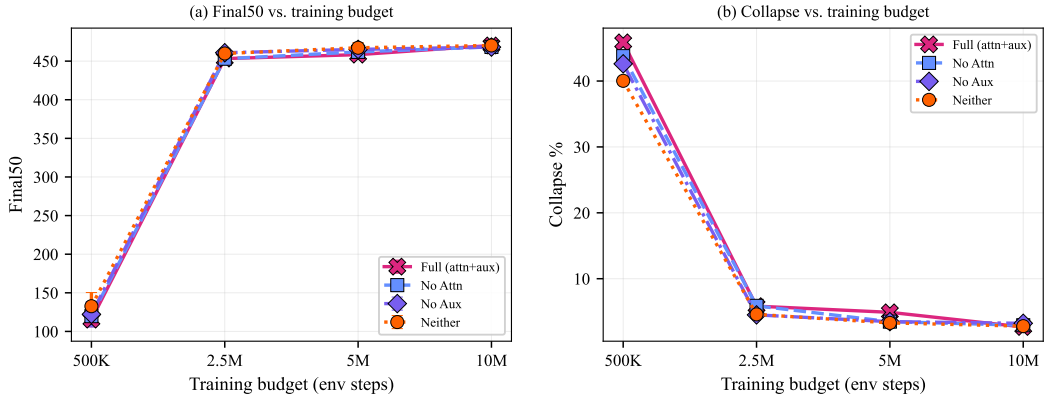

Figure 7: Sample-efficiency curves. **(a)** All configurations converge to the same regime ($\sim$470) by 10M steps. **(b)** Collapse drops from $\sim$45% at 500K to $<$6% at 2.5M+, confirming the training-budget artifact. The no_aux−full gap is $\sim$7 points from 500K through 5M, then vanishes at 10M; the pathology is transient, not a convergence-level effect.

Table 8: Sample-efficiency: Final50 mean $\pm$ std (5 seeds) at each budget.

Config	500K	2.5M	5M	10M
Full	115.3 ± 9.3	453.3 ± 2.6	458.4 ± 4.5	470.3 ± 4.1
No Attn	119.8 ± 8.8	453.1 ± 5.6	461.9 ± 6.5	468.4 ± 5.1
No Aux	122.2 ± 16.5	460.8 ± 5.6	465.2 ± 5.4	468.2 ± 4.0
Neither	132.6 ± 17.7	459.9 ± 3.4	467.3 ± 6.8	470.3 ± 2.5

F Architecture Sweep

We test the pathology across 6 architecture variants by swapping recurrence (GRU [8], LSTM [17], or feedforward) and attention (cross-attention, self-attention, additive [1], or mean pooling) independently while keeping all other hyperparameters fixed. Each configuration is run with and without the auxiliary loss (5 seeds each, 60 runs total). A MAAC-style variant places the auxiliary head on the centralized critic instead of the actor.

Table 9: Architecture sweep on Overcooked AA (25K episodes, 5 seeds each). Cohen's d: positive means aux hurts. The pathology generalizes across recurrence types and attention mechanisms but is absent with mean pooling and reverses when aux targets the critic.

Recurrence	**Attention**	**Aux loc.**	**+aux Final50**	**−aux Final50**	d	**Pathology?**
LSTM	cross-attn	actor	436.5 ± 22.1	454.0 ± 21.2	$+0.80$	**yes**
GRU	additive	actor	440.4 ± 21.8	524.8 ± 116.5	$+1.01$	**yes**
GRU	self-attn	actor	527.6 ± 145.2	544.4 ± 156.1	$+0.11$	no[‡]
GRU	mean pool	actor	431.7 ± 25.3	422.8 ± 33.2	-0.30	no
FF (none)	cross-attn	actor	588.1 ± 195.3	609.6 ± 208.7	$+0.11$	no[‡]
GRU	cross-attn	**critic**	546.6 ± 156.6	458.1 ± 21.5	-0.79	**reversed**

[‡]Both conditions exhibit high intrinsic cross-seed variance ($\sigma > 145$) in both the +aux and −aux cells, making the aux effect undetectable.

The pathology appears whenever *any* recurrence (GRU or LSTM) is paired with *any* learned attention mechanism (cross-attention or additive) and a constant auxiliary loss on the actor. Mean pooling, which has no learned attention parameters, is immune ($d = -0.30$). The MAAC-style critic-auxiliary configuration reverses the effect ($d = -0.79$): the auxiliary gradient flows through the critic rather than the actor's belief encoder, so it corrupts value estimates (which are recomputed each epoch) rather than the policy representation (which accumulates damage through recurrence).

G Hyperparameters

Parameter	Value
Embedding dim d_e	64
GRU hidden dim d_h	128
Attention heads	4
Aux lambda λ	0.05 (constant, unless annealed)
PPO [36] clip ϵ	0.2
PPO epochs	10
GAE λ [35]	0.95
Discount γ	0.99
Learning rate	5×10^{-4} (Adam [21], ϵ=10^{-5})
Gradient clip	10.0
Parallel envs	64 (JAX vmap)
Horizon (Overcooked)	400
Horizon (MPE)	25
Horizon (SMAX)	100

Table 10: Shared hyperparameters across all VABL configurations and baselines.

H Figure Accessibility

All figures in this paper are designed for accessibility across viewing conditions. One author has color vision deficiency, which motivated a systematic approach to figure design:

- **Color palette.** We use the IBM colorblind-safe palette (magenta, blue, purple, orange, gold, teal), which maintains perceptual distinctness under deuteranopia, protanopia, and tritanopia. The pathological configuration (Full VABL) uses magenta throughout all figures for consistent identification.
- **Redundant encoding.** Every data series is distinguished by at least two independent visual channels. Dot plots use distinct marker shapes (cross, square, diamond, circle). Bar charts use hatching patterns (diagonal, dots, backslash, crosshatch) in addition to color. Line plots use distinct line styles (solid, dashed, dash-dot, dotted) with large markers.

- **Grayscale readability.** All figures remain interpretable when printed in black-and-white. We verified this by converting each figure to grayscale and confirming that all data series remain distinguishable through shape, pattern, and line style alone.
- **Font consistency.** Figure text (axis labels, legends, annotations) uses the same serif font family as the paper body to maintain visual coherence. We avoid matplotlib's default sans-serif rendering.
- **No duplicate titles.** Figures use LaTeX captions only, with no redundant matplotlib titles baked into the image. This avoids visual clutter and ensures consistent typography.

We encourage the MARL community to adopt similar accessibility practices. Approximately 8% of males and 0.5% of females have some form of color vision deficiency, and many readers access papers in printed grayscale. Relying solely on color to distinguish data series excludes these readers.

NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: The abstract and introduction state five contributions, each supported by specific experimental evidence (Section 6) and theoretical analysis (Section 4). Scope limitations are explicitly discussed: the pathology is environment-dependent (present on Overcooked, absent on MPE) and specific to co-learning non-stationarity (absent in supervised CIFAR-100).

2. **Limitations**

   Answer: [Yes]

   Justification: Section 8 discusses environment scope (Overcooked + MPE, not SMAC/Hanabi), theoretical simplifications (quadratic loss, i.i.d. noise assumptions in Proposition 1), the cross-domain negative result, and the fact that VABL is not state-of-the-art in absolute reward.

3. **Theory assumptions and proofs**

   Answer: [Yes]

   Justification: Proposition 1 states all assumptions (quadratic loss near convergence, i.i.d. noise, spectral radius condition). A proof sketch appears in Section 4; the full proof is in Appendix A. We explicitly note that the proposition is a simplified model whose qualitative predictions we verify empirically.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: All hyperparameters are in Appendix G. Per-seed values are in Appendix C. The codebase uses JAX with fixed seeds; all runs are deterministic given the same seed and hardware. We plan to release code and data upon acceptance.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: We will release the full codebase (JAX implementations of all 7 methods, training scripts, analysis scripts), all raw result JSONs ($>$200 runs), and a one-command reproduction script. An anonymized repository will be provided during review.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Section 5 specifies environments, training budgets, number of seeds, vectorization setup, and evaluation metrics. Appendix G provides all hyperparameters. Appendix E provides multi-budget curves.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: All MARL experiments use 5 random seeds. We report population standard deviation throughout and Cohen's d for effect sizes. Per-seed values are in Appendix C. The vision cross-domain test uses 1 seed (noted as a limitation).

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: All experiments run on a single NVIDIA RTX 5090 (32GB). Per-run wall times are $\sim$8.5 minutes for 10M-step Overcooked runs and $\sim$27 minutes for 100K-episode MPE runs, with 64 vectorized environments via JAX vmap. Total compute across all experiments: $\sim$200 runs $\times$ $\sim$10 minutes $\approx$ 33 GPU-hours. The vision experiment uses a separate RTX 3060 Ti for smoke testing.

9. **Code of ethics**

   Answer: [Yes]

Justification: This work studies optimization dynamics in cooperative multi-agent systems using standard benchmarks (Overcooked, MPE, CIFAR-100). No human subjects, private data, or dual-use concerns apply.

10. **Broader impacts**

    Answer: [N/A]

    Justification: This is foundational research on optimization pathologies in multi-agent reinforcement learning. The findings are methodological (how to design loss functions) and do not have direct negative societal applications. The practical recommendation (anneal or stop-gradient auxiliary losses) improves training reliability without introducing new risks.

11. **Safeguards**

    Answer: [N/A]

    Justification: The paper does not release pre-trained models, scraped datasets, or generative systems. The released code trains small cooperative agents on game benchmarks with no misuse potential.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: We use Overcooked-AI (MIT license, Carroll et al. 2019), JaxMARL (Apache 2.0), CIFAR-100 (research use), and standard JAX/Flax/PyTorch libraries. All are properly cited.

13. **New assets**

    Answer: [Yes]

    Justification: We release a benchmark dataset of >200 training runs with raw episode rewards, and JAX implementations of 7 cooperative MARL methods. Documentation and reproduction instructions will accompany the release.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: No human subjects or crowdsourcing involved.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Answer: [N/A]

    Justification: No human subjects involved.

16. **Declaration of LLM usage**

    Answer: [N/A]

    Justification: LLMs were not used as a component of the core methodology. LLMs were used for writing assistance and code generation during the research process, which does not require declaration per NeurIPS policy.